

- 5 (a)** graph: straight line at constant potential = V_0 from $x = 0$ to $x = r$ B1
 curve with decreasing gradient M1
 passing through $(2r, 0.50V_0)$ and $(4r, 0.25V_0)$ A1 [3]
- (b)** graph: straight line at $E = 0$ from $x = 0$ to $x = r$ B1
 curve with decreasing gradient from (r, E_0) M1
 passing through $(2r, \frac{1}{4}E_0)$ A1 [3]
(for 3rd mark line must be drawn to $x = 4r$ and must not touch x-axis)